PROBLEM CL14

Gas-Turbine Diffuser Module: Exit State, Force and Torque

Original statement · CL14.pdf

Stator | Swirl removal | Angular momentum | Perfect gas

OFFICIAL PROBLEM STATEMENT

The diffuser module at the exit of a gas turbine consists of a **fixed** aerofoil cascade (stator), of inner and outer radii R_i and R_o, respectively, together with a conical diffuser, as depicted in the figure of CL14.pdf. The gas enters the module with velocity field $\bar{v}_1 = u_1\bar{e}_z + u_\theta\bar{e}_\theta$, having both axial and azimuthal components, pressure p_1 and density ρ_1, and leaves the module with a purely axial velocity field $\bar{v}_2 = u_2\bar{e}_z$, pressure p_2 and density ρ_2. For the analysis of the problem, one may introduce the following assumptions:

- The effect of gravity is entirely negligible.
- Flow properties at the entrance and exit sections are uniform.
- The air around the casing is at rest with uniform pressure p_a.
- The working gas behaves as a perfect gas (gas constant R_g, specific heat ratio γ).
- All the inner walls are adiabatic.

With the above simplifications, write the equations that determine the values of ρ_2 and u_2, as well as the net force, F, and torque, M, acting on the diffuser module, in terms of the known quantities (p_a, p_1, ρ_1, p_2, u_1, u_θ, R_i, R_o, γ and R_g).

Source: CL14.pdf (pages 1-4, quiz of Nov 5, 2012, with typed solution). Notes.pdf, Chapter 3: integral mass balance Eq. 3.11, integral momentum balance Eq. 3.36, integral angular momentum balance Eq. 3.38, integral total energy balance Eq. 3.46.

Worked solution (4-phase method)

PHASE 1

Hypotheses, Data and Unknowns

- Steady, axisymmetric; neglect $\bar{g}$; uniform properties at both sections; perfect gas, $h = c_pT = \frac{\gamma}{\gamma-1}\frac{p}{\rho}$; adiabatic walls; the module is fixed (no work).
- Inlet: annulus $R_i < r < R_o$ (area $A_1 = \pi(R_o^2 - R_i^2)$). Exit: full disc $r < R_o$ (area $A_2 = \pi R_o^2$).

- The ambient air at p_a touches only the outer cylindrical casing, whose normal is radial: by axisymmetry it gives no axial force and no axial torque ($\bar{F}_{ext} = 0$, $\bar{M}_{ext} = 0$). Absolute pressures are therefore used, as in the official solution (the gauge form would wrongly add $p_a \pi R_i^2$ on the hub face, which is not in contact with the ambient air).
- Data: $[p_a,\ p_1,\ \rho_1,\ p_2,\ u_1,\ u_\theta,\ R_i,\ R_o,\ \gamma,\ R_g]$. Unknowns: $[\rho_2,\ u_2,\ F,\ M]$.

Symbol	Meaning	SI unit
u_1, u_θ	axial and azimuthal inlet velocity	m/s
p_1, ρ_1; p_2, ρ_2	inlet and exit states	Pa, kg/m^3
u_2	exit axial velocity	m/s
R_i, R_o	inner and outer radii	m
F, M	axial force and torque on the module	N, N·m

PHASE 2

Frames, Control Volume and Surfaces

- Frame fixed to the module; right-handed cylindrical basis $(\bar{e}_r, \bar{e}_\theta, \bar{e}_z)$, $\bar{e}_r \wedge \bar{e}_\theta = \bar{e}_z$, with $\bar{e}_z$ along the flow; origin on the axis.
- Surfaces (as in the official solution): Σ_{in} (inlet annulus, $\bar{n} = -\bar{e}_z$), Σ_{out} (exit disc, $\bar{n} = +\bar{e}_z$), Σ_w (inner wall of the casing and outer wall of the cone) and Σ_a (surfaces around the stator aerofoils). All are fixed.

$$\boxed{F\,\bar{e}_z = \int_{\Sigma_w,\Sigma_a} \left[p\,\bar{n} - \bar{\bar{\tau}}' \cdot \bar{n}\right] d\sigma + \underbrace{\bar{F}_{ext}}_{0}}$$

$$\boxed{M\,\bar{e}_z = \int_{\Sigma_w,\Sigma_a} \bar{x} \wedge \left[p\,\bar{n} - \bar{\bar{\tau}}' \cdot \bar{n}\right] d\sigma + \underbrace{\bar{M}_{ext}}_{0}}$$

PHASE 3

Balances, Surface by Surface

Mass (Notes Eq. 3.11)

$$\underbrace{\cancel{\frac{d}{dt}\int_{V_c}\rho\,dV}}_{0\ \text{(steady)}} + \int_{\Sigma_c}\rho\,(\bar{v}-\underbrace{\bar{v}_c}_{0})\cdot\bar{n}\,d\sigma = 0$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\,\bar{v}\cdot\bar{n}\,d\sigma$
Σ_{in}	$-\rho_1 u_1 \pi (R_o^2 - R_i^2)$
Σ_{out}	$+\rho_2 u_2 \pi R_o^2$
$\Sigma_w,\ \Sigma_a$	0

$$\boxed{\rho_1 u_1 \pi (R_o^2 - R_i^2) = \rho_2 u_2 \pi R_o^2 = G} \qquad (1)$$

Linear momentum, axial component (Notes Eq. 3.36)

$$\underbrace{\cancel{\frac{d}{dt}\int_{V_c}\rho\bar{v}\,dV}}_{0\ \text{(steady)}} + \int_{\Sigma_c}\rho\bar{v}\,[(\bar{v}-\underbrace{\bar{v}_c}_{0})\cdot\bar{n}]\,d\sigma = -\int_{\Sigma_c} p\,\bar{n}\,d\sigma + \int_{\Sigma_c}\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma + \underbrace{\cancel{\int_{V_c}\rho\bar{g}\,dV}}_{\text{neglect } \bar{g}}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho v_z\,(\bar{v}\cdot\bar{n})\,d\sigma$	$-\int p\,n_z\,d\sigma$	$\int(\bar{\bar{\tau}}'\cdot\bar{n})_z\,d\sigma$
Σ_{in}	$-\rho_1 u_1^2 \pi (R_o^2 - R_i^2)$	$+p_1 \pi (R_o^2 - R_i^2)$	0 (uniform)
Σ_{out}	$+\rho_2 u_2^2 \pi R_o^2$	$-p_2 \pi R_o^2$	0 (uniform)
$\Sigma_w,\ \Sigma_a$	0	$-\int p\,n_z\,d\sigma$	$\int(\bar{\bar{\tau}}'\cdot\bar{n})_z\,d\sigma$

Boxed terms: the force of the walls on the gas, $-F$ (with $\bar{F}_{ext} = 0$). The swirl gives no axial momentum flux.

$$\boxed{F = \pi(R_o^2 - R_i^2)\left(p_1 + \rho_1 u_1^2\right) - \pi R_o^2\left(p_2 + \rho_2 u_2^2\right)} \tag{2}$$

Angular momentum, axial component (Notes Eq. 3.38)

$$\underbrace{\frac{d}{dt}\int_{V_c} \rho\,\bar{x}\wedge\bar{v}\,dV}_{0} + \int_{\Sigma_c} \rho\,(\bar{x}\wedge\bar{v})(\bar{v}\cdot\bar{n})\,d\sigma = -\int_{\Sigma_c} \bar{x}\wedge p\,\bar{n}\,d\sigma + \int_{\Sigma_c} \bar{x}\wedge(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\,(\bar{x}\wedge\bar{v})_z\,(\bar{v}\cdot\bar{n})\,d\sigma$	pressure and viscous moments
Σ_{in}	$\int_{R_i}^{R_o} \rho_1\,(r u_\theta)(-u_1)\,2\pi r\,dr = -\frac{2\pi}{3}\rho_1 u_1 u_\theta (R_o^3 - R_i^3)$	0 ($\bar{x}\wedge\bar{e}_z$ has no z component)
Σ_{out}	0 (no swirl)	0
$\Sigma_w,\ \Sigma_a$	0	$-M$

On Σ_{in}, $\bar{x} = r\bar{e}_r + z\bar{e}_z$ and the z component of $\bar{x}\wedge(u_1\bar{e}_z + u_\theta\bar{e}_\theta)$ is $r u_\theta$ (from $\bar{e}_r\wedge\bar{e}_\theta = \bar{e}_z$). The antiderivative of r^2 is $r^3/3$; Barrow's rule between R_i and R_o gives $(R_o^3 - R_i^3)/3$.

$$\boxed{M = \frac{2\pi}{3}\,\rho_1 u_1 u_\theta\left(R_o^3 - R_i^3\right)} \tag{3}$$

Total energy (Notes Eq. 3.46)

$$\underbrace{\frac{d}{dt}\int_{V_c} \rho\left(e + \tfrac{v^2}{2}\right)dV}_{0\ \text{(steady)}} + \int_{\Sigma_c} \rho\left(e + \frac{v^2}{2}\right)\left[(\bar{v} - \underbrace{\bar{v}_c}_{0})\cdot\bar{n}\right]d\sigma = -\int_{\Sigma_c} p\,\bar{v}\cdot\bar{n}\,d\sigma + \int_{\Sigma_c} \bar{v}\cdot(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma + \underbrace{\int_{V_c}\rho\bar{g}\cdot\bar{v}\,dV}_{\text{neglect } \bar{g}} - \int_{\Sigma_c}\bar{q}\cdot\bar{n}\,d\sigma + \underbrace{\int_{V_c} Q_R\,dV}_{0}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho \left(e + \frac{v^2}{2}\right)(\bar{v}\cdot\bar{n})\,d\sigma$	$-\int p\,\bar{v}\cdot\bar{n}\,d\sigma$	$\int \bar{v}\cdot(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma$	$-\int \bar{q}\cdot\bar{n}\,d\sigma$
Σ_{in}	$-G\left(e_1 + \frac{u_1^2+u_\theta^2}{2}\right)$	$+p_1 u_1 \pi (R_o^2 - R_i^2)$	0	0
Σ_{out}	$+G\left(e_2 + \frac{u_2^2}{2}\right)$	$-p_2 u_2 \pi R_o^2$	0	0
$\Sigma_w,\ \Sigma_a$	0	0 ($\bar{v}=0$)	0 ($\bar{v}=0$)	0 (adiabatic)

$$\boxed{\frac{\gamma}{\gamma-1}\frac{p_1}{\rho_1} + \frac{u_1^2+u_\theta^2}{2} = \frac{\gamma}{\gamma-1}\frac{p_2}{\rho_2} + \frac{u_2^2}{2}} \qquad (4)$$

(1) and (4) give ρ_2 and u_2; then (2) gives F and (3) gives M directly; $T_2 = p_2/(\rho_2 R_g)$ from the equation of state.

Discrepancy with the official solution (sign of the torque)

CL14.pdf uses $\bar{x}\wedge\bar{e}_\theta = -r\bar{e}_z$ and obtains $M = -\frac{2\pi}{3}\rho_1 u_1 u_\theta (R_e^3 - R_i^3)$ (it calls the outer radius R_e). With the right-handed basis $(\bar{e}_r, \bar{e}_\theta, \bar{e}_z)$ one has $\bar{e}_r \wedge \bar{e}_\theta = +\bar{e}_z$, so $\bar{x}\wedge\bar{e}_\theta = +r\bar{e}_z$ and the sign is the one in (3). The magnitude is the same; physically, the torque of the gas on the stator has the sense of the incoming swirl, since the stator removes it. The computed sign is kept.

PHASE 4

Interpretation, Limits and Dimensional Check

- (4): the stagnation enthalpy is conserved; the kinetic energy of the swirl, $u_\theta^2/2$, is recovered as enthalpy (pressure and temperature) downstream.
- For $u_\theta = 0$ there is nothing to remove and $M = 0$. M is the torque that the mounting of the stator must withstand.
- Dimensions (SI): $[(p+\rho u^2)R^2] = \mathrm{N}$; $[\rho u_1 u_\theta R^3] = \mathrm{N\cdot m}$.
- Numerical illustration (hypothetical data, extension): $p_1 = 1.2\times10^5$ Pa, $\rho_1 = 1.1$ kg/m^3, $u_1 = 150$ m/s, $u_\theta = 80$ m/s, $p_2 = 1.3\times10^5$ Pa, $R_i = 0.2$ m, $R_o = 0.5$ m, $\gamma = 1.4$: eliminating $\rho_2 = G/(\pi R_o^2 u_2)$ in (4) gives a quadratic in u_2 with positive root $u_2 = 118.57$ m/s, $\rho_2 = 1.1689$ kg/m^3; then $F = -19512$ N (the module is pushed upstream by the pressure rise) and $M = 3234.6$ N·m.